

C Programming

1. What is the output of the following C code?

<!-- end list -->

```
``c
```

```
int a = 5;
```

```
printf("%d", a++ * 2);
```

```
``
```

A) 10

B) 11

C) 12

D) 13

Answer: A) 10

2. Which header file is essential for using input/output functions like printf() and scanf()?

A) \<string.h\>

B) \<stdlib.h\>

C) \<stdio.h\>

D) \<math.h\>

Answer: C) \<stdio.h\>

3. What is the size of an int data type in a typical 32-bit system?

A) 1 byte

B) 2 bytes

C) 4 bytes

D) 8 bytes

Answer: C) 4 bytes

4. The entry point for the execution of a C program is the function:

- A) start()
- B) main()
- C) run()
- D) execute()

Answer: B) main()

5. Which of the following is NOT a valid C variable name?

- A) total_sum
- B) _count
- C) 4th_value
- D) sum2

Answer: C) 4th_value

6. What is the correct way to declare a constant integer value in C?

- A) const int PI = 3.14;
- B) integer PI = 3.14;
- C) \#define PI 3.14
- D) constant PI = 3.14;

Answer: C) \#define PI 3.14

7. Which operator is used for finding the remainder of a division?

- A) /
- B) %
- C) \#
- D) \$

Answer: B) %

8. What is the purpose of the break statement in a C program?

- A) To exit the program immediately.

- B) To skip the current iteration of a loop.
- C) To terminate the execution of a loop or switch statement.
- D) To continue with the next statement outside the loop.

Answer: C) To terminate the execution of a loop or switch statement.

9. How many times will the following for loop execute?

<!-- end list -->

```
``c
for (int i = 0; i < 5; i++);
``
```

- A) 4
- B) 5
- C) 6
- D) Infinite times

Answer: B) 5

10. Which storage class is used to declare global variables?

- A) auto
- B) register
- C) static
- D) extern

Answer: D) extern

11. What is a pointer in C?

- A) A variable that stores the address of another variable.
- B) A variable that stores a floating-point number.
- C) A variable that stores the value of a constant.
- D) A data type for characters.

Answer: A) A variable that stores the address of another variable.

12. The operator used to get the address of a variable is:

- A) * (Asterisk)
- B) & (Ampersand)
- C) -\> (Arrow)
- D) . (Dot)

Answer: B) & (Ampersand)

13. Which function is used to dynamically allocate a single block of requested memory?

- A) realloc()
- B) free()
- C) calloc()
- D) malloc()

Answer: D) malloc()

14. Which of the following is a string in C?

- A) An array of integers.
- B) An array of characters terminated by a null character (`\\0`).
- C) A pointer to a floating-point number.
- D) A single character enclosed in single quotes.

Answer: B) An array of characters terminated by a null character (`\\0`).

15. Which function is used to calculate the length of a string?

- A) strlen()
- B) size()
- C) length()
- D) strlen()

Answer: D) strlen()

16. What is the value of x after the following operations?

<!-- end list -->

```
```c
```

```
int x = 10;
```

```
x %= 3;
```

```
```
```

A) 1

B) 0

C) 3

D) 7

Answer: A) 1

17. In a switch statement, if no case matches the expression, which block is executed?

A) The first case.

B) The last case.

C) The default case.

D) The program stops.

Answer: C) The default case.

18. What is the primary difference between a while loop and a do-while loop?

A) A while loop checks the condition at the end; a do-while loop checks at the beginning.

B) A do-while loop executes the body at least once, even if the condition is false.

C) A while loop is used for definite iterations; a do-while loop for indefinite.

D) There is no functional difference; only syntax differs.

Answer: B) A do-while loop executes the body at least once, even if the condition is false.

19. Which keyword is used to return from a function?

A) return

B) exit

C) done

D) goto

Answer: A) return

20. What is the data type of the expression `5 > 3`?

A) int (0 or 1)

B) bool

C) char

D) float

Answer: A) int (0 or 1)

21. What is the term for a function that calls itself?

A) Inline function

B) Nested function

C) Recursive function

D) Standard function

Answer: C) Recursive function

22. Which is the correct way to access the third element of an array named `arr`?

A) `arr(3)`

B) `arr[2]`

C) `arr[3]`

D) `arr{2}`

Answer: B) `arr[2]`

23. What does the expression `*ptr` do if `ptr` is a pointer?

A) Gets the address stored in `ptr`.

B) Gets the value stored at the address pointed to by `ptr`.

C) Increments the address in `ptr`.

D) Multiplies the value by a random number.

Answer: B) Gets the value stored at the address pointed to by `ptr`.

24. What is a struct in C?

- A) A user-defined data type that stores a collection of items of the same type.
- B) A built-in data type for complex numbers.
- C) A user-defined data type that groups variables of different data types.
- D) A reserved keyword for defining global constants.

Answer: C) A user-defined data type that groups variables of different data types.

25. The format specifier %f is used for which data type in scanf() or printf()?

- A) char
- B) long int
- C) double
- D) float

Answer: D) float

26. What is the result of the bitwise AND operation: 5 & 3? (5 is 0101, 3 is 0011)

- A) 7
- B) 1
- C) 5
- D) 3

Answer: B) 1

27. The continue statement in a loop:

- A) Terminates the loop.
- B) Skips the rest of the loop's current iteration and proceeds to the next iteration.
- C) Skips the entire loop.
- D) Continues to the statement immediately following the loop.

Answer: B) Skips the rest of the loop's current iteration and proceeds to the next iteration.

28. Which storage class has a lifetime throughout the execution of the program and is initialized to zero by default?

- A) auto
- B) register
- C) static
- D) extern

Answer: C) static

29. Which C standard library function is used to copy one string into another?

- A) strcpy()
- B) strcpv()
- C) copy_str()
- D) strncpy()

Answer: B) strcpv()

30. The operator ?: is known as the:

- A) Unary operator
- B) Conditional operator (Ternary)
- C) Logical operator
- D) Bitwise operator

Answer: B) Conditional operator (Ternary)

31. What is the output of the following C code?

<!-- end list -->

```
```c
```

```
int x = 1, y = 0;
```

```
printf("%d", x && y);
```

```
```
```

- A) 0
- B) 1

- C) Error
- D) Undefined

Answer: A) 0

32. What is the maximum number of dimensions an array can have in C?

- A) 2
- B) 3
- C) Any number
- D) Depends on the compiler and memory

Answer: D) Depends on the compiler and memory

33. What is the output of `sizeof(char)`?

- A) 1
- B) 2
- C) 4
- D) 8

Answer: A) 1

34. Which mode should be used to open a file for writing, where the content is added to the end of the existing file?

- A) "w"
- B) "r"
- C) "a"
- D) "w+"

Answer: C) "a"

35. The `void` keyword is used for:

- A) Specifying a function that does not return any value.
- B) Specifying a function that accepts no arguments.
- C) Specifying a generic pointer type.
- D) All of the above.

Answer: D) All of the above.

36. What is the correct format specifier for a pointer address?

A) %p

B) %u

C) %x

D) %s

Answer: A) %p

37. Which function is used to free the dynamically allocated memory?

A) delete()

B) clear()

C) free()

D) deallocate()

Answer: C) free()

38. What does the term lvalue refer to?

A) A value that can be placed on the left side of an assignment operator.

B) A value that can only be placed on the right side of an assignment operator.

C) A logical value (True/False).

D) A label in a goto statement.

Answer: A) A value that can be placed on the left side of an assignment operator.

39. What is the value of i after the loop terminates?

<!-- end list -->

```
```c
```

```
int i;
```

```
for (i = 0; i < 10; i++) {}
```

```
```
```

A) 9

B) 10

C) 11

D) Undefined

Answer: B) 10

40. In C, arrays are passed to functions by:

A) Value

B) Reference

C) Copy

D) Name

Answer: B) Reference

41. Which operator has the lowest precedence?

A) +

B) *

C) =

D) ||

Answer: C) =

42. What is the output of the following code?

<!-- end list -->

```
```c
```

```
printf("%d", 10 || 5);
```

```
```
```

A) 1

B) 0

C) 10

D) 5

Answer: A) 1

43. Which loop is best suited for an iteration where the number of repetitions is known in advance?

A) while

B) do-while

C) for

D) goto

Answer: C) for

44. The enum keyword is used to create:

A) New data types that store lists of integer constants.

B) New data types that can hold different data types.

C) A collection of functions.

D) A list of floating-point numbers.

Answer: A) New data types that store lists of integer constants.

45. What is the escape sequence for a newline character?

A) \\t

B) \\r

C) \\n

D) \\

Answer: C) \\n

46. Which function is used to compare two strings?

A) strncmp()

B) compare()

C) strcmp()

D) equalstr()

Answer: C) strcmp()

47. What does the preprocessor directive `\#include` do?

- A) Includes a function definition.
- B) Includes a function call.
- C) Includes the content of another file into the current file.
- D) Includes a global variable declaration.

Answer: C) Includes the content of another file into the current file.

48. What is the output of `sizeof(int \*)`?

- A) 2
- B) 4
- C) 8
- D) Depends on the architecture (usually 4 or 8)

Answer: D) Depends on the architecture (usually 4 or 8)

49. What is the default initial value of a local variable with the auto storage class?

- A) 0
- B) 1
- C) A garbage value
- D) -1

Answer: C) A garbage value

50. What is the purpose of the `typedef` keyword?

- A) To create new functions.
- B) To define a new data structure.
- C) To provide a synonym (alias) for an existing data type.
- D) To change the default data type of an identifier.

Answer: C) To provide a synonym (alias) for an existing data type.

51. The process of converting a high-level language program into machine language is known as:

- A) Linking

- B) Debugging
- C) Compilation
- D) Execution

Answer: C) Compilation

52. Which format specifier is used to print a single character?

- A) %c
- B) %d
- C) %s
- D) %f

Answer: A) %c

53. What is the correct way to define a union in C?

- A) union Data { int i; float f; };
- B) union Data [int i; float f;];
- C) Union Data { int i; float f; };
- D) typedef union { int i; float f; } Data;

Answer: A) union Data { int i; float f; };

54. The difference between a structure and a union is that:

- A) Structures store all members simultaneously; a union stores only one member at a time.
- B) Structures are dynamic; unions are static.
- C) Structures can only have integer members; unions can have any type.
- D) Structures are defined with typedef; unions are not.

Answer: A) Structures store all members simultaneously; a union stores only one member at a time.

55. Which function is used to write a formatted string to a file?

- A) printf()
- B) fwrite()
- C) fputc()

D) fprintf()

Answer: D) fprintf()

56. The \#pragma directive is a type of:

A) Compiler command

B) Preprocessor directive

C) Linking command

D) Runtime instruction

Answer: B) Preprocessor directive

57. What is the output of the following C code?

<!-- end list -->

```
``c
```

```
int i = 0;
```

```
while (i++ < 3) {
```

```
    printf("%d", i);
```

```
}
```

```
...
```

A) 012

B) 123

C) 1234

D) 0123

Answer: B) 123

58. What is the correct declaration for a function that takes an integer and returns a float?

A) int function(float x);

B) float function(int x);

C) void function(int x);

D) float function(void);

Answer: B) float function(int x);

59. Which of the following is an example of an unary operator?

A) +

B) /

C) --

D) &&

Answer: C) --

60. What is the range of values for a signed char (assuming 8-bit)?

A) 0 to 255

B) -128 to 127

C) -32768 to 32767

D) 0 to 65535

Answer: B) -128 to 127

61. What is the main use of the register storage class?

A) To store global variables.

B) To request the compiler to store the variable in a CPU register for faster access.

C) To define variables that retain their value between function calls.

D) To automatically initialize variables to zero.

Answer: B) To request the compiler to store the variable in a CPU register for faster access.

62. Which function is used to reverse a string?

A) strrev()

B) reverse()

C) strrrvse()

D) reversestr()

Answer: A) strrev()

63. The null pointer is represented by:

- A) NULL
- B) \\0
- C) NUL
- D) ''

Answer: A) NULL

64. What is type casting in C?

- A) Defining a new data type.
- B) Changing the data type of a variable during its declaration.
- C) Explicitly converting a value from one data type to another.
- D) Implicitly converting a value from one data type to another.

Answer: C) Explicitly converting a value from one data type to another.

65. The sizeof operator returns the size of its operand in:

- A) bits
- B) words
- C) bytes
- D) kilobytes

Answer: C) bytes

66. What is the correct way to call a function named calculate that takes two integers?

- A) calculate(10, 20);
- B) call calculate(10, 20);
- C) function calculate(10, 20);
- D) calculate 10, 20;

Answer: A) calculate(10, 20);

67. Which of the following is not a standard library function for string manipulation?

- A) strcmp
- B) strcat

C) strstr

D) strcat

Answer: C) strstr

68. Which operator has higher precedence: * (multiplication) or + (addition)?

A) * (multiplication)

B) + (addition)

C) They have the same precedence

D) Depends on the operands

Answer: A) * (multiplication)

69. What is the purpose of the rewind() function in file handling?

A) To delete the file.

B) To close the file.

C) To move the file pointer to the beginning of the file.

D) To move the file pointer to the end of the file.

Answer: C) To move the file pointer to the beginning of the file.

70. Which of the following data types is typically 8 bytes in size?

A) int

B) float

C) long long int

D) short int

Answer: C) long long int

71. What is the use of comma operator in C?

A) To perform multiplication.

B) To separate function arguments.

C) To combine multiple expressions into one, where the whole expression's value is the value of the rightmost expression.

D) To define an array.

Answer: C) To combine multiple expressions into one, where the whole expression's value is the value of the rightmost expression.

72. The term dangling pointer refers to:

- A) A pointer that has not been initialized.
- B) A pointer that points to a memory location that has been deallocated.
- C) A pointer that points to an array.
- D) A pointer to a function.

Answer: B) A pointer that points to a memory location that has been deallocated.

73. Which function is used to read a single character from the console?

- A) puts()
- B) getc()
- C) getch()
- D) gets()

Answer: C) getch()

74. In C, all preprocessor directives begin with which symbol?

- A) \$
- B) @
- C) \#
- D) &

Answer: C) \#

75. What is the output of the following code snippet?

<!-- end list -->

```
``c
```

```
int a = 10;
```

```
a = a << 1;
```

```
printf("%d", a);
```

```
...
```

A) 5

B) 10

C) 20

D) 1

Answer: C) 20

76. An array name without an index in a C expression often decays into:

A) The address of the last element.

B) The size of the array.

C) A pointer to the first element.

D) A local variable.

Answer: C) A pointer to the first element.

77. The two primary types of linking are:

A) Hard and Soft

B) Internal and External

C) Static and Dynamic

D) Primary and Secondary

Answer: C) Static and Dynamic

78. What does the static keyword do when applied to a local variable inside a function?

A) Makes it a global variable.

B) Allows it to retain its value between function calls.

C) Hides it from external files.

D) Initialises it to a garbage value.

Answer: B) Allows it to retain its value between function calls.

79. The heap is used for:

- A) Storing local variables.
- B) Storing global variables.
- C) Storing dynamically allocated memory.
- D) Storing function calls.

Answer: C) Storing dynamically allocated memory.

80. Which of the following is a valid float constant?

- A) 100
- B) 3.14F
- C) 'c'
- D) "hello"

Answer: B) 3.14F

81. What is the difference between `\\0` and `0` in C?

- A) `\\0` is a character constant (ASCII 0); `0` is an integer constant.
- B) They are exactly the same.
- C) `\\0` is used for numbers; `0` is used for characters.
- D) `\\0` is the null pointer; `0` is a boolean false.

Answer: A) `\\0` is a character constant (ASCII 0); `0` is an integer constant.

82. The function `main` without any return type explicitly specified will implicitly return:

- A) `char`
- B) `void`
- C) `int`
- D) `float`

Answer: C) `int`

83. The process of passing a copy of the argument's value to the function is called:

- A) Call by Reference
- B) Call by Value
- C) Call by Address

D) Call by Pointer

Answer: B) Call by Value

84. Which header file contains general utility functions including memory management functions?

A) `<stdio.h>`

B) `<stdlib.h>`

C) `<string.h>`

D) `<math.h>`

Answer: B) `<stdlib.h>`

85. What does the term pass by value mean in function calls?

A) The function can modify the original variable.

B) The function receives a copy of the argument's address.

C) The function receives a copy of the argument's value.

D) The function always returns a value.

Answer: C) The function receives a copy of the argument's value.

86. In conditional compilation, which directive is used to check if a macro is NOT defined?

A) `#ifdef`

B) `#ifndef`

C) `#if`

D) `#undef`

Answer: B) `#ifndef`

87. The standard input stream in C is known as:

A) `stderr`

B) `stdout`

C) `stdin`

D) `stdall`

Answer: C) `stdin`

88. What is the size of the array declared as `int arr[5] = {1, 2, 3};`?

- A) 3 bytes
- B) 5 bytes
- C) 12 bytes
- D) 20 bytes

Answer: D) 20 bytes (assuming 4 bytes per int)

89. What is the output of the following C code?

<!-- end list -->

```
``c
int x = 5;
x = ++x * 2;
printf("%d", x);
...

```

- A) 10
- B) 11
- C) 12
- D) 18

Answer: C) 12

90. Which logical operator is used for OR operation?

- A) &
- B) |
- C) &&
- D) ||

Answer: D) ||

91. What is a null statement in C?

- A) A statement that returns a NULL value.
- B) A statement that has no effect, written as a single semicolon (;).
- C) A statement that initializes a pointer to NULL.
- D) A statement that includes the void keyword.

Answer: B) A statement that has no effect, written as a single semicolon (;).

92. The malloc() function returns a pointer of type:

- A) int *
- B) void *
- C) char *
- D) float *

Answer: B) void *

93. What is the term for the variables that are declared outside of all functions?

- A) Local variables
- B) Global variables
- C) Auto variables
- D) Register variables

Answer: B) Global variables

94. Which of the following is a correct way to comment out a single line in C?

- A) \# This is a comment
- B) // This is a comment
- C) /* This is a comment
- D) ** This is a comment

Answer: B) // This is a comment

95. What is the output of the following C code?

<!-- end list -->


```
``c  
printf("%c", 65);  
``
```

- A) 65
- B) 'A'
- C) 'a'
- D) Error

Answer: B) 'A'

96. The purpose of const keyword is to:

- A) Define a constant variable that cannot be changed.
- B) Define a new data type.
- C) Define a static variable.
- D) Define a volatile variable.

Answer: A) Define a constant variable that cannot be changed.

97. Which of the following is an example of an escape sequence?

- A) sum
- B) main
- C) \\t
- D) printf

Answer: C) \\t

98. What is the correct way to declare an array of 5 pointers to integers?

- A) int *arr[5];
- B) int arr[5]*;
- C) int (*arr)[5];
- D) *int arr[5];

Answer: A) int *arr[5];

99. Which file handling function is used to read a formatted string from a file?

A) fscanf()

B) fprintf()

C) fgets()

D) fputs()

Answer: A) fscanf()

100. What is the term for a block of code that performs a specific task and can be reused?

A) Loop

B) Array

C) Header

D) Function

Answer: D) Function